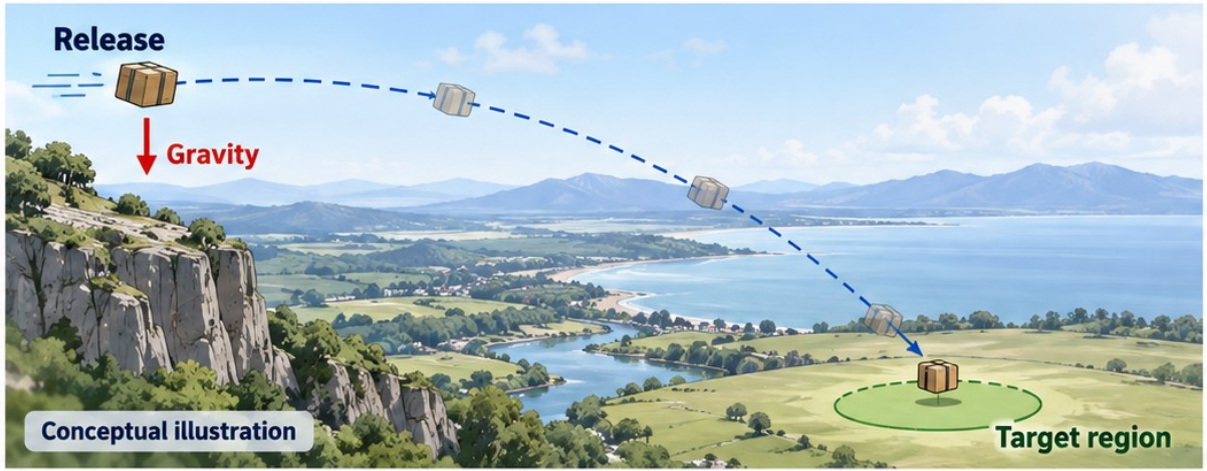
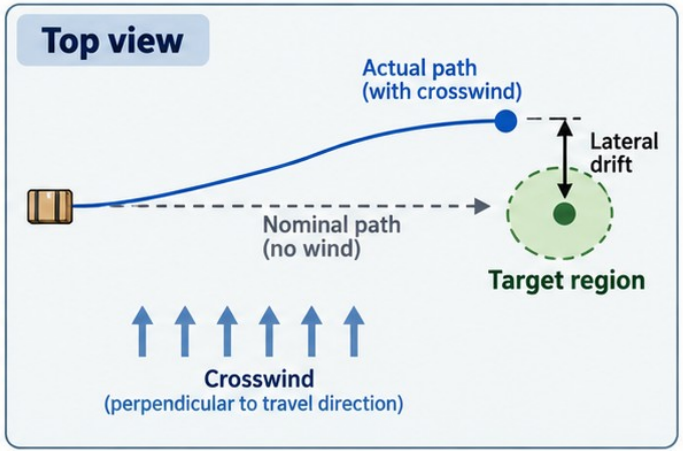
Project 01

Projectile Delivery in Crosswind with Quadratic Drag





Scenario + objective

A released package must reach a target region. Investigate how crosswind and air resistance change its flight and landing position.

Physical reasoning

Gravity acts downward. Quadratic drag opposes velocity relative to the air. Crosswind can cause lateral drift.

Information needed

Release position (m), velocity (m/s), mass (kg), wind vector (m/s), drag properties and target geometry. Separate supplied data from assumptions.

Scope decisions

Is steady wind adequate? Is 3D motion needed? Can the package be treated as a point? Lift, spin and gusts are optional.

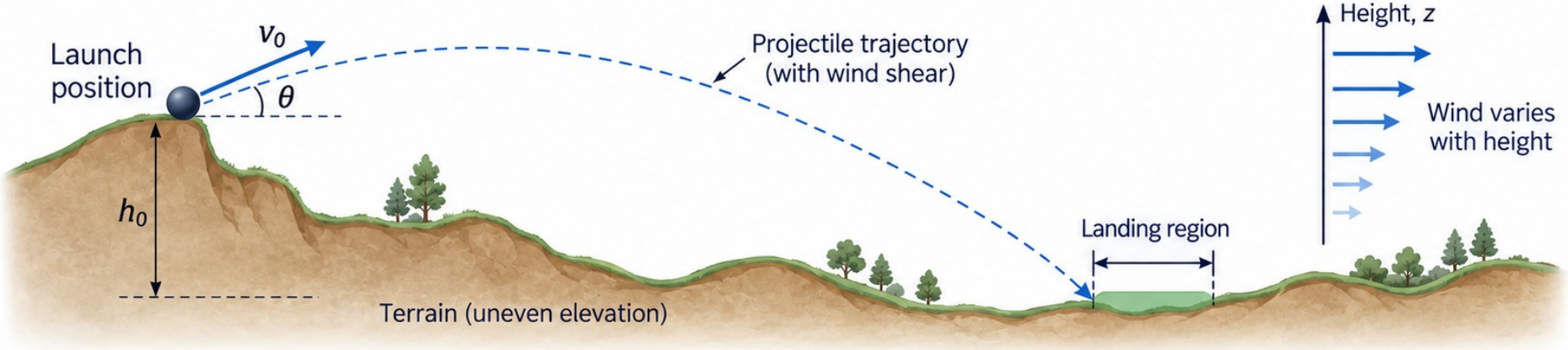
Possible evidence

Compare trajectories and landing displacement across wind cases. Use a no-drag reference and a timestep check where appropriate.

Explain delivery accuracy and justify important omissions.

Project 02

Projectile Launch from Elevated Terrain under Wind Shear



Scenario + objective

An object launches above an uneven landing surface while wind changes with height. Investigate whether it reaches the acceptable landing region.

Physical reasoning

The object encounters different airflow at different heights. Flight ends at intersection with the terrain, which need not be at zero elevation.

Information needed

Launch position and terrain elevations (m), launch velocity (m/s), wind observations/profile (m/s versus height in m), object properties and acceptable landing region.

Scope decisions

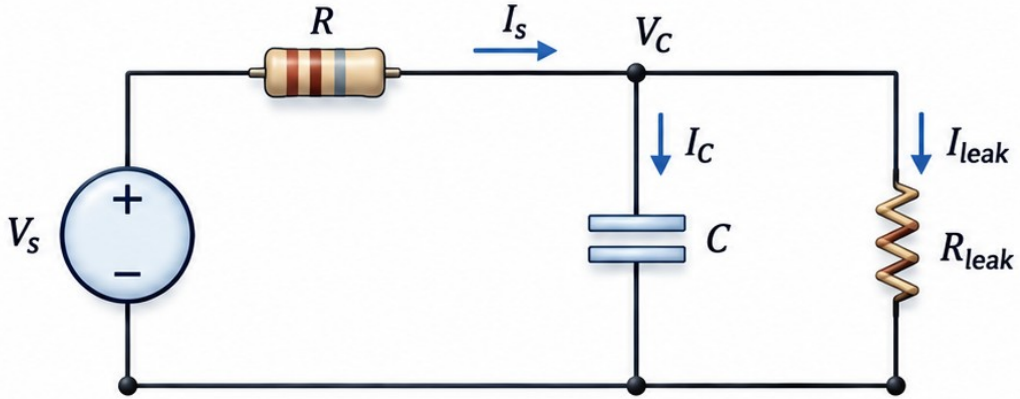
Which terrain detail matters? Is a vertical-plane model defensible? How should the model handle heights beyond the supported wind information? Gusts and obstacles remain optional.

Possible evidence

Possible evidence includes trajectory over terrain, landing location and impact speed, comparison with uniform wind, and a simpler terrain reference. Explain sensitivity to wind-profile or elevation assumptions.

Project 03

RC Charging with Leakage and Temperature-Dependent Components





Temperature cases

Use supplied or justified changes in *R*, C or leakage.

Do not assume every parameter changes in the same direction.

Scenario + objective

A capacitor or sensor node charges while leakage and temperature alter its response. Investigate the change from an ideal RC baseline.

Physical reasoning

Source current splits into capacitor current and leakage current. Temperature may change *R*, C or leakage; use supported dependencies.

Information needed

Supply and initial voltage (V); *R* and leakage resistance (Ω); C (F); temperature range (°C or K); component-temperature information.

Scope decisions

Which temperature dependence matters? Is one leakage path enough? Fixed-temperature cases are a useful start; self-heating is optional.

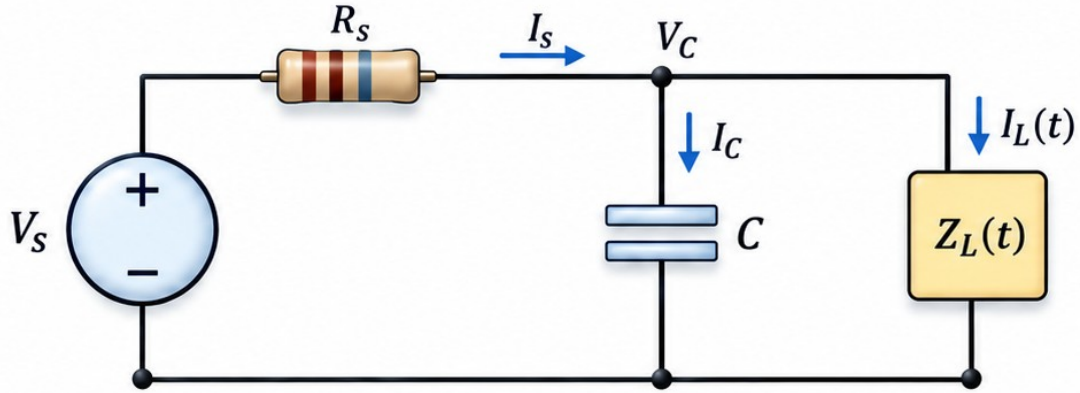
Possible evidence

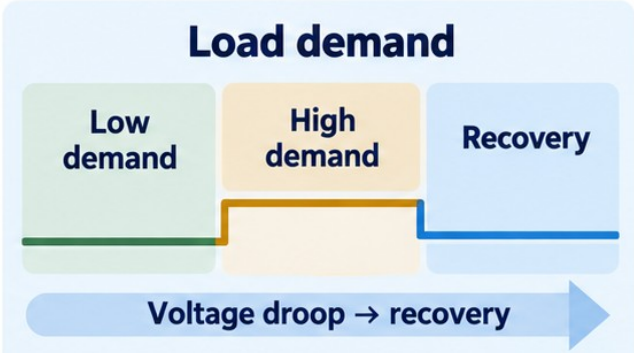
Compare voltage histories, charging time and final voltage. Use a no-leakage reference and check long-time behaviour.

Explain the effect on the sensor node and justify important omissions.

Project 04

RC Sensor Circuit with Internal Source Resistance and Time-Varying Load





SCENARIO + OBJECTIVE

A non-ideal source supplies an RC sensor node whose load changes with time. Investigate voltage droop, recovery and acceptable operation.

PHYSICAL REASONING

Internal resistance creates a current-dependent voltage loss. The capacitor stores charge and influences the transient response as load demand changes.

INFORMATION NEEDED

Source voltage (V), internal resistance (ohm), capacitance (F), initial node voltage (V), load schedule (A or ohm versus s), and an acceptable voltage band if supplied.

SCOPE DECISIONS

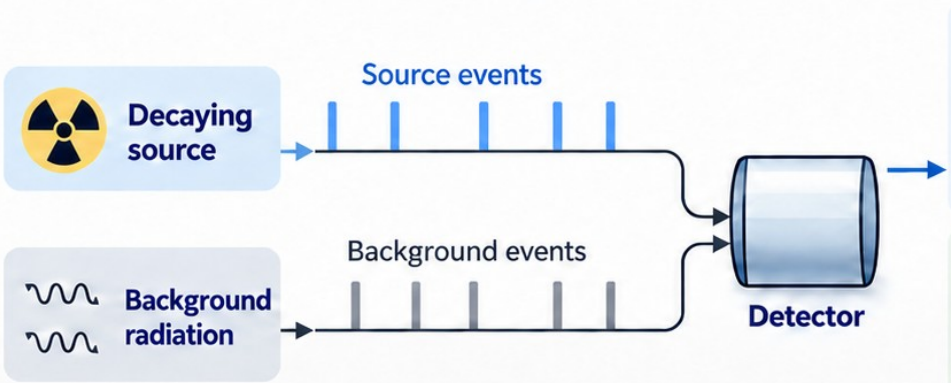
Represent the load as a current demand or resistance consistently. Are transients important? Is constant source resistance adequate? Keep thermal and wiring effects optional.

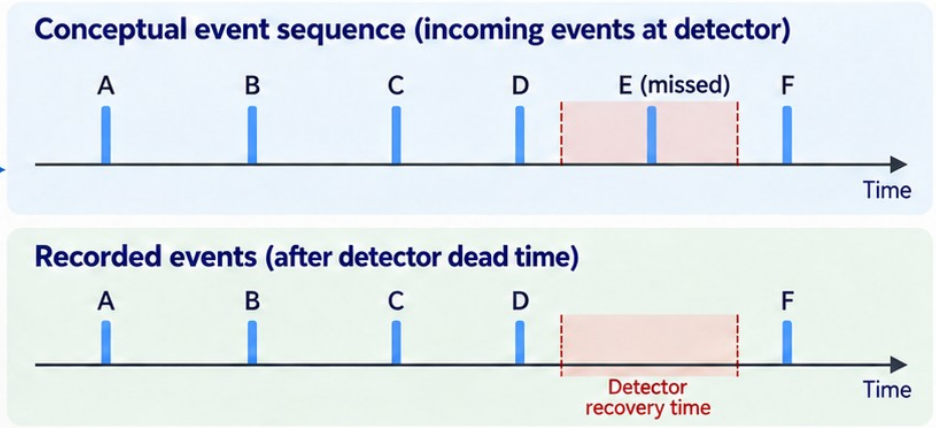
POSSIBLE EVIDENCE

Compare aligned load and node-voltage histories, minimum voltage and recovery behaviour. Use a constant-load or ideal-source case as a reference and check a steady segment. State whether the supported operating band is maintained.

Project 05

Radioactive Decay with Background Radiation and Detector Dead Time





Scenario + objective

A detector records decaying-source events together with background, while recovery time can prevent some events being counted. Investigate how recorded counts differ from the source behaviour.

Physical reasoning

Background adds events; dead time can remove recorded events at high rates. Counts, count rate and source activity are distinct quantities.

Information needed

Source/background counts, acquisition interval (s), count rate (counts/s), decay information, detector dead time (s), and efficiency/calibration when available.

Scope decisions

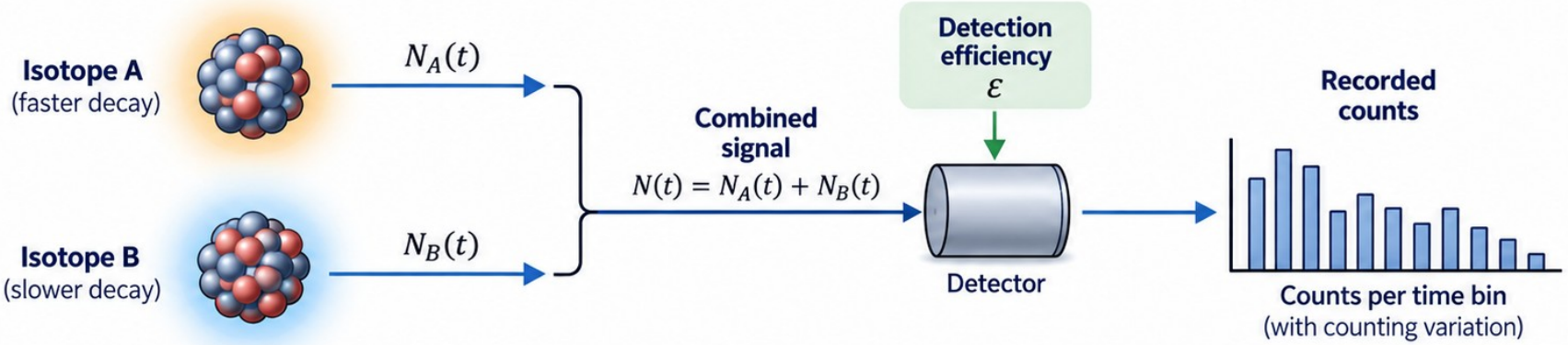
How is background established? Which dead-time model is supported? Is efficiency adequately constant? Do not add pile-up or detailed detector physics without evidence.

Possible evidence

Possible evidence includes expected and recorded rate histories, the effect of background and event rate, and zero-dead-time or low-rate checks. Explain the regimes where detector effects materially change the conclusion.

Project 06

Two-Isotope Decay Mixture with Efficiency and Counting Noise



Scenario + objective

Recorded counts arise from two decaying components with detection efficiency and statistical variation. Investigate whether the observed time history supports distinguishing the components.

Physical reasoning

Faster and slower decays contribute to a combined signal. The observation window, relative contributions and counting variation affect what can be inferred.

Information needed

Decay timescales (s or another consistent time unit), initial contributions/activity where supplied, bin width (s), counts per bin, efficiency (dimensionless), and background information if used.

Scope decisions

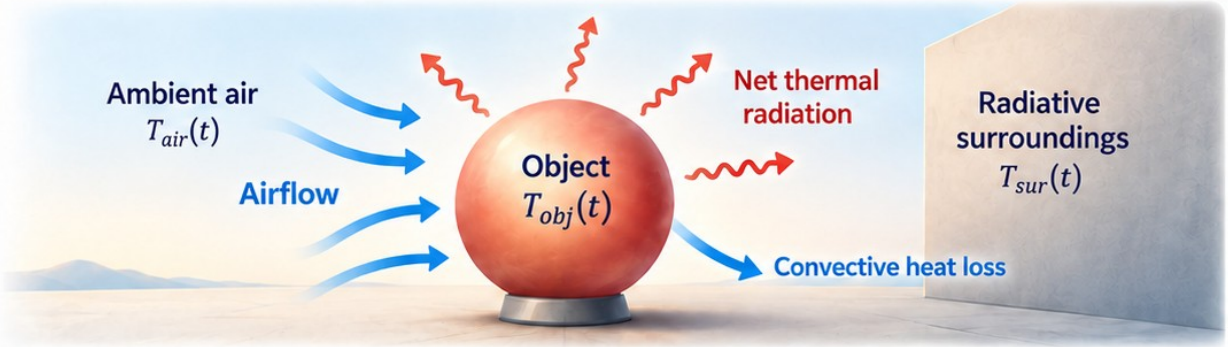
Are timescales sufficiently distinct? Is the observation window informative? Is efficiency constant? Advanced parameter estimation is optional; do not require identification from inadequate data.

Possible evidence

Possible evidence includes separate expected components, their total and noisy recorded counts; a one-component reference and repeated simulations can support checking. Explain why noise or similar decay times can make the mixture ambiguous.

Project 07

Cooling with Thermal Radiation and Changing Ambient Temperature







Temperature aspects

Treat the ambient air temperature and the radiative-surroundings temperature as separate quantities if they differ.

Radiation calculations require absolute temperature (K).

Scenario + objective

A warm object exchanges heat with its surroundings while ambient temperature changes. Investigate how these effects alter its temperature history.

Physical reasoning

Convection and net thermal radiation contribute to energy transfer. Distinguish ambient air from radiative surroundings if their temperatures differ; radiation calculations require absolute temperature.

Information needed

Initial and ambient temperature histories (°C or K), radiative-surroundings temperature if separate, mass (kg), heat capacity (J/kg/K), area (m^2), convection coefficient ($W/m^2/K$), emissivity and time (s).

Scope decisions

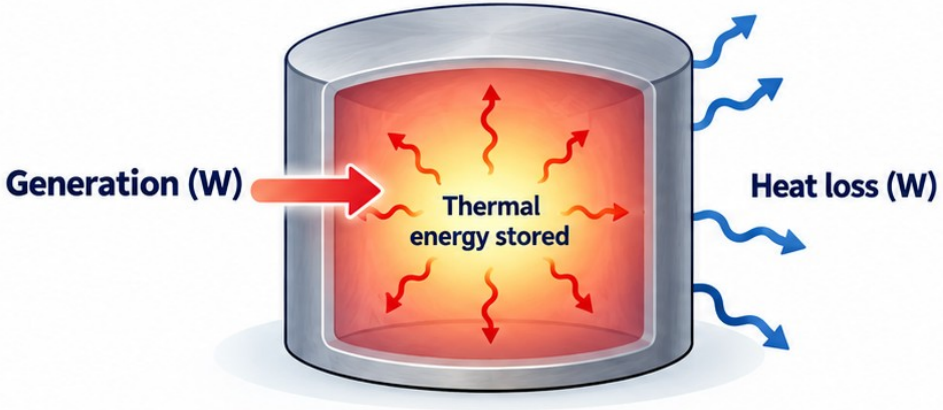
Is one object temperature adequate? Is a constant convection coefficient reasonable? Are ambient variation and radiation material? Spatial heat-flow modelling is optional.

Possible evidence

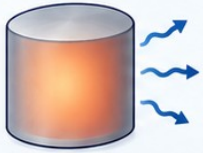
Possible evidence includes object and ambient temperature histories, comparison with fixed surroundings or convection-only behaviour, and an equilibrium/no-temperature-difference check. Explain which exchange mechanism matters over the supported range.

Project 08

Cooling of a Heated Object with Internal Heat Generation

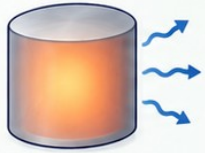


Generation < loss — cooling



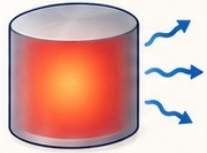
Temperature decreases towards a lower value.

Generation = loss — steady



Temperature remains constant at a steady value (usually above ambient if generation is nonzero).

Generation > loss — warming



Temperature increases towards a higher value until a new balance is reached.

Scenario + objective

An initially heated object continues generating heat while losing energy to its surroundings. Investigate conditions for cooling, warming or steady temperature.

Physical reasoning

Temperature follows the balance between heat generation and heat loss. Internal generation can sustain a steady temperature above ambient.

Information needed

Generation power or schedule (W versus s), initial/ambient temperature, thermal capacity (J/K) or mass and specific heat, geometry and supported heat-loss information.

Scope decisions

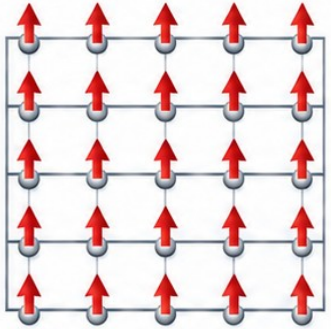
Is generation uniform and prescribed? Is a lumped temperature adequate? Keep spatial gradients, changing material properties and multiple heat-transfer paths optional unless needed.

Possible evidence

Possible evidence includes temperature and generation histories, the effect of one generation change, and comparison with zero generation. Check whether heat generation balances loss in an apparent steady state; explain the resulting operating condition.

Project 09
Bounded 2D Ising Ferromagnet

Low temperature

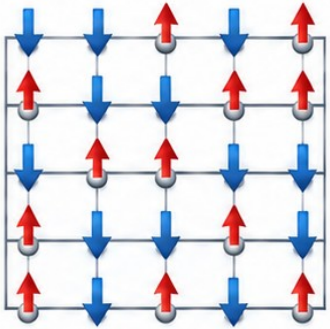


Conceptual configurations (5 × 5 lattice)

Spin up (+1)

Spin down (–1)

High temperature



Physical random updates
Use a supplied physical sampling rule that depends on the energy and temperature.

Equilibrate
Run updates to reach a representative state.

Measure
Calculate observables such as magnetisation and energy.

Scenario + objective

A finite lattice of interacting up/down spins provides a simple ferromagnetic model. Investigate how temperature changes domains, magnetisation and energy.

Physical reasoning

Neighbour interactions and temperature influence spin configurations. Random updates belong to a supplied physical sampling rule; they are not arbitrary flips.

Information needed

Lattice size, interaction/ energy scale, consistently defined temperature or reduced temperature, boundary condition, initial state, update count and random seed. Use a supplied implementation scaffold.

Scope decisions

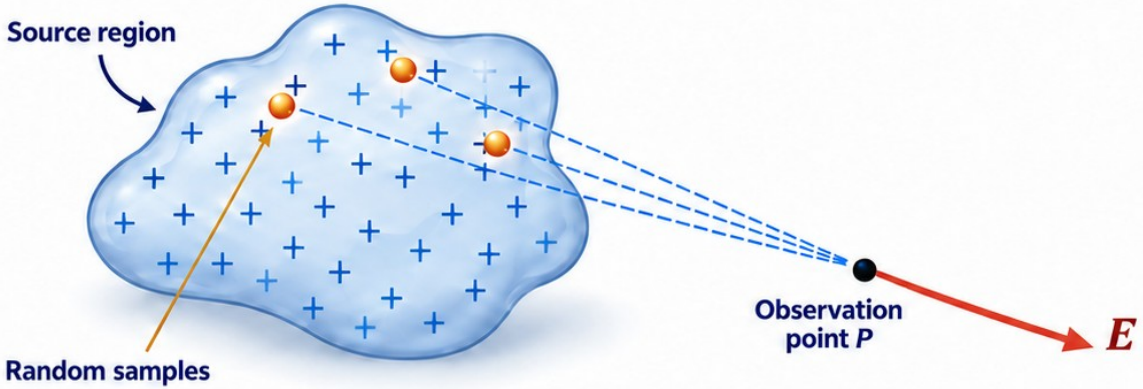
How long should the system equilibrate? Do initial state and finite size matter? Fix most settings and vary one principal factor; algorithm derivation and precise transition claims are beyond Core.

Possible evidence

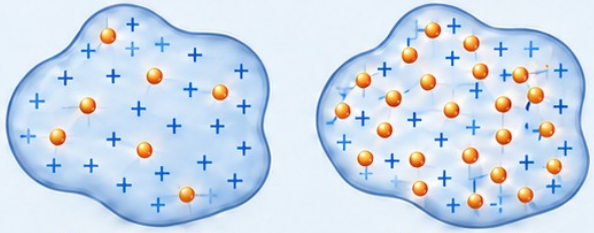
Possible evidence includes comparable lattice snapshots and averaged magnetisation/ energy after equilibration, repeated seeds and longer-run checks. Explain finite-run variation and the limits of claims from a small lattice.

Project 10

Monte Carlo Field of an Irregular Charged Body



Conceptual random sampling of the source



Fewer samples (conceptual)

More samples (conceptual)

SCENARIO + OBJECTIVE

An irregular charged body has a specified charge distribution. Use physically meaningful random samples to estimate the electric field at selected observation points.

PHYSICAL REASONING

Vector contributions from charge elements add to the net electric field. Sampling and charge weighting must represent the stated geometry and density.

INFORMATION NEEDED

Geometry and observation coordinates (m), volume or surface charge density with consistent units (C/m^3 or C/m^2), sample count, random seed and any reference geometry supplied.

SCOPE DECISIONS

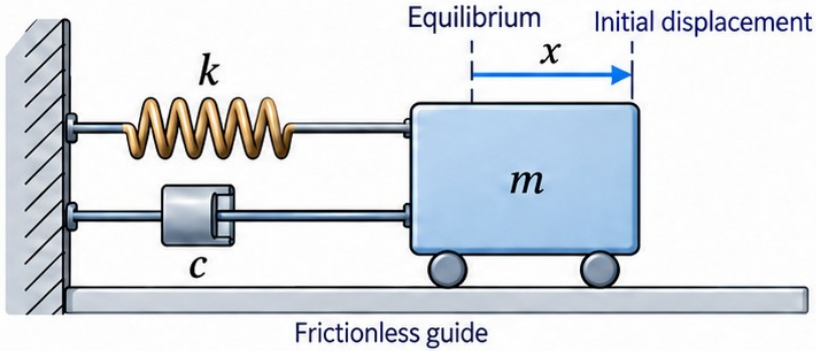
Is the source a volume or a thin surface? Does the sampling represent it without bias? Begin with external observation points; near-source or singular cases require explicit treatment and remain optional.

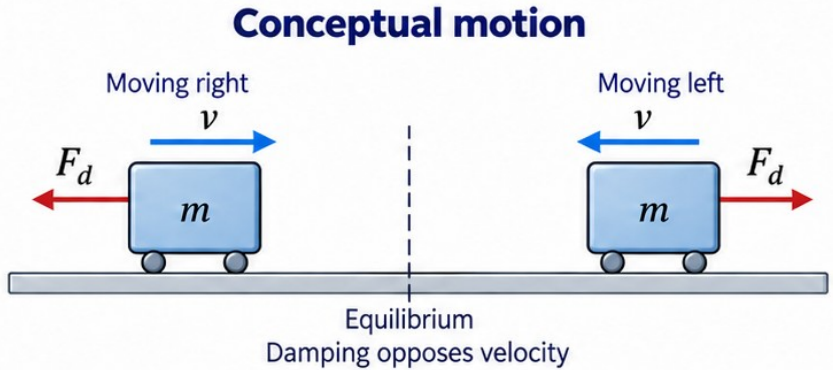
POSSIBLE EVIDENCE

Possible evidence includes field components/magnitude (N/C), direction at observation points, repeated-run spread and sample-count behaviour. A symmetric reference or far-field comparison can support validation where appropriate.

Make each random sample physically meaningful; report spread and a reproducible check.

Project 11
Damped Mass–Spring Response





Scenario + objective

A disturbed mass–spring system loses energy as it moves. Investigate how damping changes its transient response and settling behaviour.

Physical reasoning

The spring restores displacement; damping opposes motion and dissipates mechanical energy. Use an initial disturbance as the simplest baseline; additional forcing requires an explicit scope decision.

Information needed

Mass (kg), stiffness (N/m), damping parameter with units matching the selected law, initial displacement (m), initial velocity (m/s), observation interval (s) and any settling criterion supplied.

Scope decisions

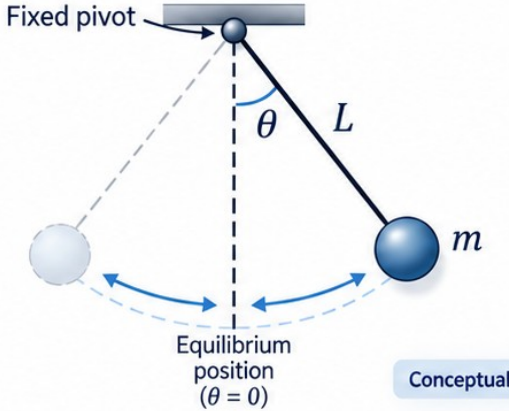
Are 1D motion, a linear spring and a simple damping law adequate? Nonlinear stiffness, changing support and continued forcing are optional.

Possible evidence

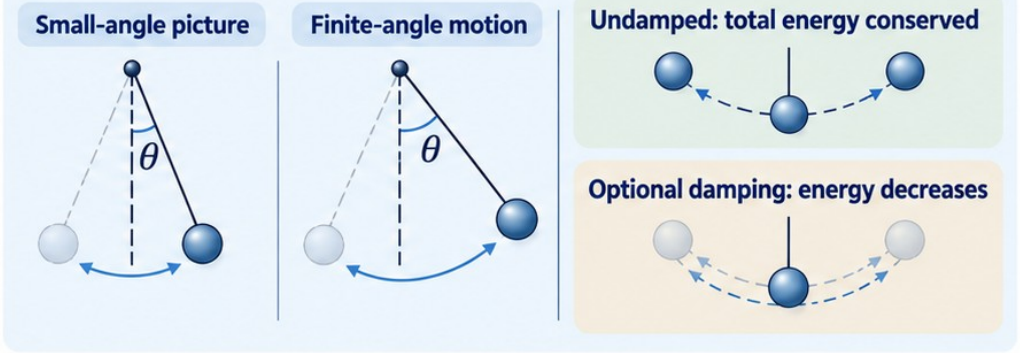
Compare displacement, energy and settling across damping cases. Use an undamped reference and refine the timestep to separate physical dissipation from numerical error.

Implement one effect that changes the response; justify important omissions.

Project 12
Finite-Angle Pendulum Motion



Conceptual illustration



Scenario + objective

A pendulum is released from an angle too large to assume small-angle motion automatically. Investigate how the response differs from the small-angle approximation.

Physical reasoning

The restoring effect depends on angle; the small-angle approximation has a limited range. Friction is optional and must be separated from numerical energy drift.

Information needed

Length (m), gravitational acceleration (m/s²), initial angle (rad), initial angular speed (rad/s), mass (kg) for energy, time (s), and supported damping information.

Scope decisions

Is a fixed-length, planar point-bob model adequate? Vary initial angle first. Pivot friction, rod inertia and moving support remain optional.

Possible evidence

Compare angular displacement and period across initial angles with a small-angle reference. Check energy and timestep behaviour; separate physical differences from numerical error.

Choose one main realism feature, validate it, and justify important omissions.